Task 1: Order Flow Imbalances (OFI)

Quantitative Research Submission

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Objective

The purpose of this task is to demonstrate the construction and interpretation of Order Flow Imbalance (OFI) features for predicting short-term returns in equity markets, as outlined in the paper *Cross-Impact of Order Flow Imbalance in Equity Markets*. Specifically, the code and analysis focus on implementing:

- Best-Level OFI
- Multi-Level OFI (Levels 1 through 10)
- Integrated OFI (via PCA)
- Cross-Asset OFI (explained, but not computable on single-symbol dataset)

Dataset

File: `first_25000_rows.csv`
Symbol(s): `AAPL (only)`
Frequency: Millisecond-level LOB snapshots
Due to the presence of only one symbol in the dataset, Cross-Asset OFI features cannot be constructed. This limitation is acknowledged and handled in the code with an explicit symbol check.

Feature Construction Summary

Best-Level OFI

Computed at Level 1 using bid and ask price and size changes based on Cont et al. (2014). Captures directional pressure from top of the book.

Multi-Level OFI

Constructed for LOB levels 1 through 10. Each level's OFI computed independently using the same logic as Best-Level.

Integrated OFI

Principal Component Analysis (PCA) was applied to the 10-dimensional OFI vector. The first principal component was retained and used as the Integrated OFI.

Cross-Asset OFI

Not implemented due to dataset constraint (only AAPL present). Code prints a clear message explaining that multi-symbol data is required.

Model Evaluation: LASSO Forecasting Summary

To evaluate OFI-based predictability, LASSO regression models were trained to forecast 1-minute, 5-minute, and 10-minute log returns.

Horizon	**R^2 Score**	**Best Alpha**	**Top Features**
log_return_1m	0.0000	0.000002	(none selected)
log_return_5m	0.0101	0.000003	ofi_level_2_lag1, ofi_level_9_lag1, ofi_level_10_lag1, ofi_level_3_lag1, ofi_level_5_lag1
log_return_10m	0.0138	0.000003	ofi_level_2_lag5, ofi_level_9_lag1, ofi_level_2_lag1, ofi_level_1_lag5, ofi_level_10_lag5

Table 1: LASSO Model Forecasting Summary

Interpretation

- **1-minute:** No predictive features selected; high-frequency noise dominates.
- **5-minute, 10-minute:** Weak but positive R^2 suggests some predictive signal in lagged multi-level OFI.
- **Signal depth:** Features from Levels 2, 9, and 10 were most prominent.

Conceptual Questions

1. Why measure OFI at multiple depth levels of the order book?

Top-of-book activity may not reflect hidden liquidity imbalances at deeper levels. By measuring OFI across levels 1 through 10, we capture broader order book pressure, which improves signal robustness and prediction of future price changes.

2. Why use LASSO instead of OLS for cross-impact estimation?

LASSO enforces sparsity, selecting only the most relevant OFI signals among many. In high-dimensional cross-impact settings (e.g., hundreds of OFIs from multiple stocks), this leads to more interpretable and stable models than OLS, which overfits and lacks regularization.

3. Why is OFI better than trade volume for short-term returns?

Trade volume aggregates all trades without directionality. OFI captures net buying/selling pressure by tracking aggressive activity on the bid vs. ask. This directional signal is empirically more predictive of near-term price movement, especially at high frequencies.

Code Repository

The full codebase and documentation for this submission are available on GitHub:
https://github.com/kaushikkumarkr/Blockhouse$_{Q}uant_{R}esearch$